

3. First-Order Deformations

3.1. For the moment, let us forget δ and compute the versal deformation, to first order, of the singularity 2.2, I or II. The computation is painful, but standard. For an arbitrary superscheme with an isolated singularity, one must:

(a) write

$$X = \text{Spec}(A), \quad A = k[X, \Theta]/\langle f_i \rangle;$$

(b) find a system of syzygies S_k ,

$$\sum_i a_i(X, \Theta) f_i = 0,$$

generating all of them modulo the trivial syzygies

$$f_i f_j - f_j f_i = 0 \quad (\text{even, or even/odd, case}),$$

$$f_i f_j + f_j f_i = 0 \quad (f_i \text{ and } f_j \text{ odd});$$

(c) compute the cohomology of the complex of free A -modules

$$\bigoplus_i \left(A \frac{\partial}{\partial X_i} \oplus A \frac{\partial}{\partial \Theta_i} \right) \xrightarrow{df/\partial X} \bigoplus_i A \frac{\partial}{\partial f_i} \xrightarrow{\text{coefficients}} \bigoplus_k A \frac{\partial}{\partial S_k}.$$

The H^1 parametrizes the deformations.

Case I. The coordinates are (z_1, z_2, θ) , the equation is

$$z_1 z_2 = 0,$$

and there is no syzygy. The universal first-order deformation is

$$z_1 z_2 = t + \tau \theta \quad (\text{over } k[t, \tau]/(t^2, t\tau)).$$

Case II is another matter.

Case II. We take the coordinates

$$z_1, z_2, \theta_1, \theta_2$$

and the equations

$$z_1 z_2 = z_1 \theta_2 = \theta_1 z_2 = \theta_1 \theta_2 = 0.$$

The syzygies are

$$\begin{cases} \theta_1(z_1 z_2) = z_1(\theta_1 z_2), \\ \theta_2(z_1 z_2) = z_2(z_1 \theta_2), \\ z_1(\theta_1 \theta_2) = \theta_1(z_1 \theta_2), \\ z_2(\theta_1 \theta_2) = -\theta_2(\theta_1 z_2). \end{cases}$$

One has to consider a system of four equations, one per syzygy, in four unknowns, one per equation, in the ring A , expressing that these unknowns

satisfy the syzygies. The task is to compute the space of solutions modulo the trivial solutions, obtained by differentiating the equations. I denote the unknowns by

$$++, \quad +-, \quad -+, \quad --.$$

The equations corresponding to the syzygies have coefficients

	++	+-	-+	--		∂_{z_1}	∂_{z_2}	∂_{θ_1}	∂_{θ_2}
1	θ_1	0	$-z_1$	0	++	z_2	z_1	0	0
2	θ_2	$-z_2$	0	0	+-	θ_2	0	0	z_1
3	0	$-\theta_1$	0	z_1	-+	0	θ_1	z_2	0
4	0	0	θ_2	z_2	--	0	0	θ_2	$-\theta_1$

each row, one syzygy, each column, one trivial solution.

Let (a, b, c, d) be a solution. Modifying it by the last two trivial solutions, one may suppose that d is even. The equations

$$z_1 d = \theta_1 b, \quad z_2 d = -\theta_2 c$$

then give $d = 0$. The linear combinations with $d = 0$ of the trivial solutions are the linear combinations of the first, the second, the third multiplied by θ_2 , and the fourth multiplied by θ_1 . This reduces us to the system

	++	+-	-+	-- = 0		1	2	$3\theta_2$	$4\theta_1$
1	θ_1	0	$-z_1$	0	++	z_2	z_1	0	0
2	θ_2	$-z_2$	0	0	+-	θ_2	0	0	$\theta_1 z_1$
3	0	$-\theta_1$	0	0	-+	0	θ_1	$\theta_2 z_2$	0
4	0	0	θ_2	0	--	0	0	0	0

Let (a, b, c) be a solution, with $d = 0$. The equation

$$\theta_1 a = z_1 c$$

imposes that, modulo (θ_2, z_2) , the pair (a, c) is a multiple of the second trivial solution. In particular

$$a \in (z_1, \theta_1, z_2, \theta_2) = (z_1, \theta_1) \oplus (z_2, \theta_2).$$

Similarly for b and c :

$$a = a_1 + a_2, \quad b = b_1 + b_2, \quad c = c_1 + c_2.$$

Modulo the first two trivial solutions, one obtains

$$a = 0, \quad b_2 = 0, \quad c_1 = 0.$$

By the last equations,

$$a_1 \in (\theta_1), \quad a_2 \in (\theta_2), \quad b_2 = c_1 = 0,$$

$$a_1 \in (\theta_1), \quad a_2 \in (\theta_2), \quad b_1 \in (\theta_1), \quad c_1 = 0, \quad b_2 = 0, \quad c_2 \in (\theta_2).$$

Finally, modulo the θ -multiples of the first trivial solutions, every solution is reduced to one of the following:

	a	b	c	d
1	θ_1	0	0	0
2	θ_2	0	0	0
3	0	θ_1	0	0
4	0	0	θ_2	0

Thus the versal first-order deformation is given by the parameters

$$t_1, t_2, \tau_1, \tau_2,$$

the coordinates

$$z_1, z_2, \theta_1, \theta_2,$$

and the equations

$$\begin{cases} z_1 z_2 + \tau_1 \theta_1 + \tau_2 \theta_2 = 0, \\ z_1 \theta_2 + t_1 \theta_1 = 0, \\ z_2 \theta_1 + t_2 \theta_2 = 0, \\ \theta_1 \theta_2 = 0. \end{cases}$$